

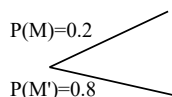
Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 8

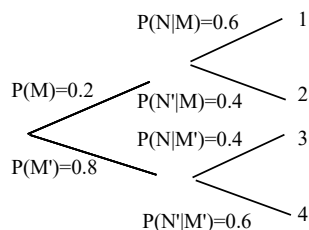
For two events, M and N, $P(M) = 0.2$, $P(N|M) = 0.6$, and $P(N|M') = 0.4$. Find $P(M|N)$.

The problem can be illustrated by a tree diagram. The first two branches of the diagram represent $P(M)$ and $P(M')$.



If $P(M) = 0.2$, $P(M') = 0.8$.

The $P(M)$ branch is divided into two further branches. The value of $P(N|M)$ is given as 0.6. The value of $P(N'|M)$, then is $1 - 0.6 = 0.4$.



The value of $P(N|M') = 0.4$.

The value of $P(N'|M') = 0.6$.

Find $P(M|N)$ by dividing the probability that both M and N occur by the probability that N occurs. Use Bayes' Theorem.

$$P(M|N) = \frac{P(M) \cdot P(N|M)}{P(M) \cdot P(N|M) + P(M') \cdot P(N|M')}$$

Branch 1 of the tree diagram shows the probability that both M and N occur.

Calculate $P(M \cap N)$ by multiplying $P(M) \cdot P(N|M)$.

$$P(M) \cdot P(N|M) = 0.12$$

Find the probability that N occurs by adding the probabilities calculated in branches 1 and 3. You have already found that the probability from branch 1 is $P(M) \cdot P(N|M) = 0.12$.

The other branch where N occurs (branch 3) is calculated by multiplying $P(M') \cdot P(N|M')$.

$$P(M') \cdot P(N|M') = 0.32$$

$$P(M|N) = \frac{0.12}{0.12 + 0.32} = \frac{0.12}{0.44} = \frac{3}{11}$$